

Query Optimization: Transformation of Relational Expressions

1 Introduction

A query can be expressed in several different ways, with different costs of evaluation. Rather than taking a relational expression as given, the optimizer considers alternative, equivalent expressions to find the least-cost evaluation plan.

Expression Equivalence

Two relational-algebra expressions are said to be **equivalent** if, on every legal database instance, the two expressions generate the same set of tuples.

- **Legal instance:** Satisfies all integrity constraints.
- **Order:** The order of tuples is irrelevant for set equivalence.
- **Multisets (SQL):** In SQL, inputs and outputs are multisets. Expressions are equivalent if they generate the same multiset of tuples.

2 Equivalence Rules

An **equivalence rule** states that expressions of two forms are equivalent. The optimizer uses these rules to transform expressions into logically equivalent alternatives.

2.1 Fundamental Rules

1. **Cascade of σ :** Conjunctive selection operations can be deconstructed into a sequence of individual selections.

$$\sigma_{\theta_1 \wedge \theta_2}(E) \equiv \sigma_{\theta_1}(\sigma_{\theta_2}(E))$$

2. **Commutativity of σ :** Selection operations are commutative.

$$\sigma_{\theta_1}(\sigma_{\theta_2}(E)) \equiv \sigma_{\theta_2}(\sigma_{\theta_1}(E))$$

3. **Cascade of Π :** In a sequence of projection operations, only the final operation is needed.

$$\Pi_{L_1}(\Pi_{L_2}(\dots(\Pi_{L_n}(E))\dots)) \equiv \Pi_{L_1}(E) \quad \text{where } L_1 \subseteq L_2 \subseteq \dots \subseteq L_n$$

4. **Selections with Joins:**

$$(a) \quad \sigma_{\theta}(E_1 \times E_2) \equiv E_1 \bowtie_{\theta} E_2$$

$$(b) \quad \sigma_{\theta_1}(E_1 \bowtie_{\theta_2} E_2) \equiv E_1 \bowtie_{\theta_1 \wedge \theta_2} E_2$$

5. **Commutativity of $\bowtie$:** Theta-join operations are commutative.

$$E_1 \bowtie_{\theta} E_2 \equiv E_2 \bowtie_{\theta} E_1$$

Note: Attribute order is ignored for equivalence.

6. **Associativity of $\bowtie$:**

- (a) Natural-join: $(E_1 \bowtie E_2) \bowtie E_3 \equiv E_1 \bowtie (E_2 \bowtie E_3)$
- (b) Theta-join: $(E_1 \bowtie_{\theta_1} E_2) \bowtie_{\theta_2 \wedge \theta_3} E_3 \equiv E_1 \bowtie_{\theta_1 \wedge \theta_3} (E_2 \bowtie_{\theta_2} E_3)$
(where θ_2 involves attributes from only E_2 and E_3)

2.2 Distribution Rules

7. **Distribution of σ over $\bowtie$:**

- (a) If θ_1 involves only attributes of E_1 :

$$\sigma_{\theta_1}(E_1 \bowtie_{\theta} E_2) \equiv (\sigma_{\theta_1}(E_1)) \bowtie_{\theta} E_2$$

- (b) If θ_1 involves only attributes of E_1 and θ_2 involves only E_2 :

$$\sigma_{\theta_1 \wedge \theta_2}(E_1 \bowtie_{\theta} E_2) \equiv (\sigma_{\theta_1}(E_1)) \bowtie_{\theta} (\sigma_{\theta_2}(E_2))$$

8. **Distribution of Π over $\bowtie$:**

- (a) $\Pi_{L_1 \cup L_2}(E_1 \bowtie_{\theta} E_2) \equiv (\Pi_{L_1}(E_1)) \bowtie_{\theta} (\Pi_{L_2}(E_2))$
- (b) More generally, including attributes L_3, L_4 needed for join condition θ :

$$\Pi_{L_1 \cup L_2}(E_1 \bowtie_{\theta} E_2) \equiv \Pi_{L_1 \cup L_2}((\Pi_{L_1 \cup L_3}(E_1)) \bowtie_{\theta} (\Pi_{L_2 \cup L_4}(E_2)))$$

9. **Commutativity of Set Operations:**

$$E_1 \cup E_2 \equiv E_2 \cup E_1 \quad \text{and} \quad E_1 \cap E_2 \equiv E_2 \cap E_1$$

Note: Set difference ($-$) is not commutative.

10. **Associativity of Set Operations:** Union and intersection are associative.

$$(E_1 \cup E_2) \cup E_3 \equiv E_1 \cup (E_2 \cup E_3)$$

11. **Distribution of σ over Set Operations:**

$$\sigma_{\theta}(E_1 \cup E_2) \equiv \sigma_{\theta}(E_1) \cup \sigma_{\theta}(E_2)$$

(Also works for $\cap$ and $-$. For $\cap$ and $-$, $\sigma_{\theta}(E_1 \cap E_2) \equiv \sigma_{\theta}(E_1) \cap \sigma_{\theta}(E_2)$ also holds).

12. **Distribution of Π over Union:**

$$\Pi_L(E_1 \cup E_2) \equiv (\Pi_L(E_1)) \cup (\Pi_L(E_2))$$

3 Special Operations and Outer Joins

13. **σ and Aggregation:** If group-by attributes G satisfy condition θ :

$$\sigma_{\theta}(G\gamma_A(E)) \equiv G\gamma_A(\sigma_{\theta}(E))$$

14. **Outer Join Commutativity:**

- Full Outer Join ($\bowtie$) is commutative.
- Left ($\bowtie_{\leftarrow}$) and Right ($\bowtie_{\rightarrow}$) are not, but: $E_1 \bowtie_{\leftarrow} E_2 \equiv E_2 \bowtie_{\rightarrow} E_1$.

15. **σ over Outer Join:** Distribution holds if θ_1 only involves attributes of one side (e.g., E_1):

$$\sigma_{\theta_1}(E_1 \bowtie_{\leftarrow} E_2) \equiv (\sigma_{\theta_1}(E_1)) \bowtie_{\leftarrow} E_2$$

16. **Outer Join to Inner Join:** If θ_1 is **null rejecting** on E_2 (evaluates to false/unknown if E_2 attributes are null):

$$\sigma_{\theta_1}(E_1 \bowtie_{\leftarrow} E_2) \equiv \sigma_{\theta_1}(E_1 \bowtie_{\rightarrow} E_2)$$

Important Warning on Outer Joins

Unlike inner joins, **outer joins are NOT associative.**

Non-Associativity of Left Outer Join

Consider $r(A, B) = \{(1, 1)\}$, $s(B, C) = \{(1, 1)\}$, and $t(A, C) = \emptyset$.

- **Case 1:** $(r \bowtie_{\leftarrow} s) \bowtie_{\leftarrow} t$
 1. $r \bowtie_{\leftarrow} s$ produces $\{(1, 1, 1)\}$.
 2. Result joined with t (empty) via left join keeps the tuple: $\{(1, 1, 1)\}$.
- **Case 2:** $r \bowtie_{\leftarrow} (s \bowtie_{\leftarrow} t)$
 1. $s \bowtie_{\leftarrow} t$ produces $\{(\text{null}, 1, 1)\}$.
 2. r left joined with this result produces $\{(1, 1, \text{null})\}$.

Since $\{(1, 1, 1)\} \neq \{(1, 1, \text{null})\}$, associativity fails.